Personalized Federated Few-Shot Learning

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***Abstract*— Personalized federated learning (PFL) learns a personalized model for each client in a decentralized manner, where each client owns private data that are not shared and data among clients are non-independent and identically distributed (i.i.d.) However, existing PFL solutions assume that clients have sufficient training samples to jointly induce personalized models. Thus, existing PFL solutions cannot perform well in a few-shot scenario, where most or all clients only have a handful of samples for training. Furthermore, existing few-shot learning (FSL) approaches typically need centralized training data; as such, these FSL methods are not applicable in decentralized scenarios. How to enable PFL with limited training samples per client is a practical but understudied problem. In this article, we propose a solution called personalized federated few-shot learning (pFedFSL) to tackle this problem. Specifically, pFedFSL learns a personalized and discriminative feature space for each client by identifying which models perform well on which clients, without exposing local data of clients to the server and other clients, and which clients should be selected for collaboration with the target client. In the learned feature spaces, each sample is made closer to samples of the same category and farther away from samples of different categories. Experimental results on four benchmark datasets demonstrate that pFedFSL outperforms competitive baselines across different settings.**

***Index Terms*— Collaboration, feature space learning, few-shot learning (FSL), non-independent and identically distributed (i.i.d.) data, personalized federated learning (PFL).**

I. Introduction

IN RECENT years, federated learning (FL) [1]–[3] has shown great promise for training a single global model over multiple clients without exposing private data with one another and thus ensuring the privacy of local data and inducing a global model in a communication-efficient manner. However, in practice, clients' local data are subjected to various distributions since they are generated on local clients with different usages or user habits, and this raises a key threat to FL: data heterogeneity [4], [5]. Data heterogeneity of different clients makes it hard to fit the entire local data with a single global model and further compromises the model performance and convergence rate.

To overcome this problem, the personalized federated learning (PFL) [6], [7] framework has been proposed to deal with the data heterogeneity of different clients. Different from typical FL solutions [1], [8], [9] that aim to induce a single shared global model by pursuing global optima of entire clients, PFL targets to jointly learn a personalized model for each participating client, with the aim of the learned local models that can fit diverse local data of clients well. Formally, let $\{c_i\}_{i=1}^n$ denote n clients that carry the same type of models but personalized by n different sets of model parameters $\{\theta_i\}_{i=1}^n$, along with n different datasets $\{\mathcal{D}_i\}_{i=1}^n$. These n datasets $\{\mathcal{D}_i\}_{i=1}^n$ are subject to n distinct distributions $\{P_i\}_{i=1}^n$, i.e., they are not independent and identically distributed (i.i.d.). For each client c_i, by defining its loss function $\mathcal{L}(\theta_i; \mathcal{D}_i)$, PFL aims at obtaining the optimal set of model parameters $\{\theta_1^*, \ldots, \theta_n^*\} = \arg\min_{\{\theta_1,\ldots,\theta_n\}} \sum_{i=1}^n \mathcal{L}(\theta_i; \mathcal{D}_i)$. Since the datasets are generated on different local clients, under different environments, usage, and context, statistical heterogeneity of data is the core issue for PFL, and most existing PFL approaches can be roughly grouped into data- and model-based approaches [10]. The former class of methods tries to alleviate the problem of weight divergence, arising from multiple rounds of local training and weight synchronization on non-i.i.d. datasets during the FL training process, by smoothing the data heterogeneity of participating clients [4], [11]–[14]. The second category of methods aims at fitting FL models to the various data distributions among clients [7], [9], [15]–[22].

However, these PFL approaches require that most or all participating clients have sufficient training data to induce the personalized models. They do not perform well in a few-shot scenario, where most or all participating clients only have few training samples for each category. Although few-shot learning (FSL) [23] has been extensively explored in diverse domains [24]–[29], most existing FSL methods typically need centralized data to induce the FSL model. To enable FSL in a decentralized scenario, a recently proposed method federated FSL with adversarial learning (FedFSL-Adv) [30] aims at inducing an FSL model in the FL setup, which can tackle new tasks with just few-shot samples. However, FedFSL-Adv still requires many-shot samples in each client to induce the FSL model and thus is not effective when applied with PFL in few-shot scenarios. We formalize this learning

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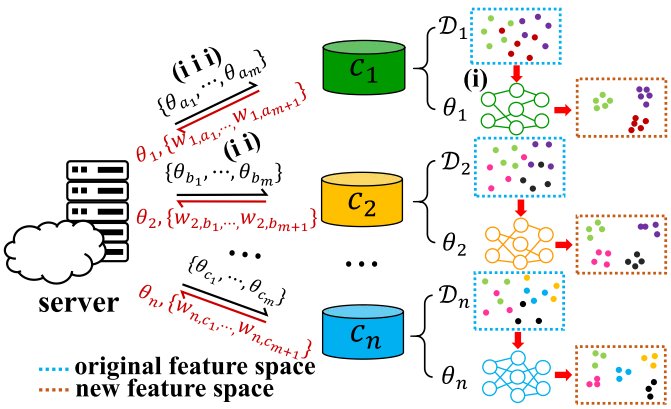

Fig. 1. Schematic framework of pFedFSL. pFedFSL learns a personalized and discriminative feature space parameterized by θ_i for each client c_i by encouraging similar clients to collaborate more with the following iterative operations: 1) a client c_i computes the aggregation weights $\{w_{i,d_1}, \ldots, w_{i,d_{m+1}}\}$ of received m models $\{\theta_{d_1}, \ldots, \theta_{d_m}\}$ and local model θ_i (where $w_{i,d_{m+1}}$ is for θ_i) by testing on its local data $\mathcal{D}_i$, gets the aggregated local model $\widetilde{\theta}_i$ by aggregating received models with computed weights, and then optimizes $\widetilde{\theta}_i$ using $\mathcal{D}_i$; 2) each client sends back the updated model and aggregation weights to the FL server; and 3) FL server decides which subset of updated local models needs to be sent to a target client for next round training.

problem as a new framework called personalized federated few-shot learning (pFedFSL). pFedFSL has many potential applications, e.g., multiple hospitals around the world want to predict some rare diseases via an FL framework and these diseases around the world are not exactly the same. The key challenges of pFedFSL are twofold: 1) the personalized model is hard to induce with scarce few-shot training samples for each client and 2) data heterogeneity exists among multiple clients and the distinct data distributions of individual clients cannot be reliably estimated with such scarce local data.

To tackle this problem, we propose a pFedFSL approach. Specifically, pFedFSL aims at learning a personalized and discriminative feature space for each client by encouraging clients that have more similar distributions to collaborate more. In these new feature spaces, each sample is made closer to samples that belong to the same category and farther away from samples that belong to other categories. Collaboration between clients with similar distributions is fostered by determining which model parameters perform well on which clients, without exposing data of local clients to the server or to other clients. The whole pFedFSL framework is shown in Fig. 1. The main contributions of our work are given as follows.

1) We focus on a practical and general PFL setting, where the training samples of each client are few-shot with a varying number of labels, and the clients' datasets are non-i.i.d. We also notice the fact that most existing FSL methods need centralized data and therefore are not applicable in a decentralized scenario. Both issues are not addressed by existing PFL solutions and FSL methods. Given that, we introduce the pFedFSL approach to induce a personalized feature space for each client, which can map the samples annotated with the same label close to each other, while the samples of different labels far apart in that space.
2) pFedFSL aims at learning a discriminative feature space for each client with few-shot training data by encouraging clients that have more similar distributions to collaborate more. This improves the performance of FL and reduces the data scarcity of individual clients. We further design a strategy to adaptively select collaborative clients to reduce the communication load and boost the training.
3) We present extensive experiments on four benchmark datasets and show that our pFedFSL outperforms standard FL strategies [1], [9], state-of-the-art PFL approaches [7], [17], [18], [31], and the competitive FedFSL-Adv method [30].

The rest of this article is organized as follows. The related work on PFL and FSL is briefly reviewed in Section II. Section III presents the technical details of our method, and Section IV reports the experimental results and analysis. Section V concludes this article and discusses future research directions.

II. Related Work

A. Personalized Federated Learning

Recently, PFL [6], [7] has emerged as a promising framework for FL with multiple clients with non-i.i.d. data. Due to its effectiveness, PFL has attracted a lot of attention, and the majority of PFL approaches can be roughly categorized into two classes: data- and model-based methods.

The goal of data-based approaches is to smooth data heterogeneity of participating clients [4], [11]–[14]. For example, Jeong *et al.* [11] proposed federated augmentation (FAug) to collectively induce a generative adversarial network (GAN) in the server. With the learned GAN model, clients can generate additional samples, which makes their local datasets becoming i.i.d. Wang *et al.* [12] proposed Favor, an experience-driven control framework, to achieve the bias caused by non-i.i.d. data by selecting a subset of the participating clients in each training round. Specifically, Favor designs a deep Q-learning formulation for client selection to maximize a reward that encourages improved model performance while penalizing the use of communication rounds.

Model-based approaches develop models that are personalized to the participating clients [7], [9], [15]–[22]. To name a few, Dinh *et al.* [17] proposed PFL with Moreau envelopes (pFedMe), where the PFL problem is formulated as a bilevel optimization problem by decoupling the training of personalized models from inducing the global model. The approach Per-FedAvg [7] finds an initial global model shared among all clients within the model-agnostic meta-learning (MAML) [32] framework and then updates the clients using their own data. Huang *et al.* [18] proposed heuristically federated attentive message passing (HeurFedAMP), which maintains a personalized cloud model for each client on the FL server and then passes the personalized models from

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clients to personalized cloud models with similar model parameters. Zhang *et al*. [31] proposed PFL using the first-order model optimization (FedFomo); this approach learns optimal weighted model combinations for each participating client by determining how much a client can benefit from models of other clients. Guo *et al*. [21] proposed PFL based on mixture of experts (PFL-MoE) to mix outputs of the personalized models and global model for achieving both personalization and generalization, while from the perspective of clustering, Duan *et al*. [22] proposed flexible clustered federated learning (FlexCFL) to group the participating clients based on the similarity between the clients' optimization directions for lower weight divergence. Besides, George *et al* [33] and George and Gurram [34] adopted distributed SGD for a set of clients to collaboratively train their individual models, in which the clients communicate only with neighorhood clients without requiring a server. The distributed SGD differs from the FL setup, where an extra server is established and clients communicate solely with this server.

Although the aforementioned approaches can induce personalized models for each client using different techniques, they cannot perform well in a more general scenario, where most or all of the participating clients have only few-shot training samples. In this scenario, clients do not have sufficient training data to induce personalized models. To enable PFL in this general setting, we propose pFedFSL, which jointly learns a personalized and discriminative feature space for each client and can prevent overfit local personalized models.

B. Few-Shot Learning

FSL [23], [35] aims at learning new concepts (class) with a handful of samples by mimicking the cognitive abilities of humans. Existing FSL problems are mainly supervised ones, and they typically can be viewed as N-way K-shot classification problems, in which the training dataset of the new task contains $K \times N$ samples from N classes, each with K examples. The core issue of FSL is the unreliability of the empirical risk minimizer, due to limited training samples. The existing approaches for FSL can be grouped into the following categories: data-driven, model-driven, and algorithm-driven [23]. Data augmentation-based FSL methods reduce the uncertainty of the empirical risk minimizer by enriching supervised information, e.g., by deriving more samples from the original few-shot samples or by using weakly labeled/unlabeled data or similar datasets [36], [37]. Model-based FSL methods constrain the hypothesis space via prior knowledge [24], [38], to reduce overfitting and make the empirical risk minimizer more reliable. With prior knowledge, algorithm-based FSL methods seek optimal model parameters by providing a good initialization or by directly learning an optimizer to output search steps [25], [32].

Unfortunately, most existing FSL methods build on the promise that the data for acquiring meta knowledge is centralized. These FSL solutions can hardly work in a decentralized manner. More recently, Fan and Huang [30] proposed the federated FSL framework and attempted it in an adversarial fashion (FedFSL-Adv). FedFSL-Adv induces an FSL model to tackle novel tasks with just a few-shot samples. However, FedFSL-Adv also builds on the premise that each participating client has sufficient training data to compose many base tasks to learn the FSL model; thus, it cannot work well in the pFedFSL scenario, in which most or all participating clients have only few-shot training samples and these data sources are non-i.i.d.

To address this problem, our proposed pFedFSL induces a personalized model for each client with limited training samples, by discovering which model parameters perform well on which clients. In this way, pFedFSL effectively handles the FSL problem with heterogeneous data sources and under the condition that data of each source are few-shot.

III. Proposed Method

A. Problem Formulation and Notation

Let $\{c_i\}_{i=1}^n$ denote a population of n clients, where each client c_i carries local model parameters θ_i^t in round t and local few-shot training data $\mathcal{D}_i = \{(\mathbf{x}_1^i, y_1^i), (\mathbf{x}_2^i, y_2^i), \ldots, (\mathbf{x}_{N_i \times K_i}^i, y_{N_i \times K_i}^i)\}$. N_i and K_i represent the number of classes and the number of training samples per class, respectively, of the ith client (a scenario often called N-way K-shot classification). For each sample $(\mathbf{x}, y)$, $\mathbf{x} \in \mathbb{R}^d$ is the d-dimensional feature vector and $y \in \{l\}_{l=1}^{N_i}$ is the ground-truth label. The set of all the samples annotated with label l in client c_i is $\mathcal{S}_l^i = \{\mathbf{x} \mid (\mathbf{x}, y) \in \mathcal{D}_i \text{ and } y = l\}$. For each participating client c_i, by defining the loss function as $\mathcal{L}(\theta_i; \mathcal{D}_i)$, we aim to learn the optimal set of personalized model parameters $\{\theta_1^*, \ldots, \theta_n^*\} = \arg\min_{\{\theta_1, \ldots, \theta_n\}} \sum_{i=1}^n \mathcal{L}(\theta_i; \mathcal{D}_i)$ that perform well on the corresponding local data via the collaboration between n clients. The framework overview of pFedFSL is shown in Fig. 1. Section II-B discusses the technical details.

B. Personalized Federated Few-Shot Learning

Metric learning [24], [39], as a main type of meta-learning [40], aims to learn a representative feature space (embedding space) and has shown great merits for tackling various FSL tasks in this new space. The learned representative feature space can embed the samples of the same class close to each other while farther away from samples of different classes, and it can also be regarded as a sort of transferable knowledge for correlated tasks. For example, pretrained visual classification networks (e.g., ResNet [41]) and natural language networks (e.g., BERT [42]) can yield representative image and language representations for related tasks, respectively. Thus, the discriminative feature space learning can also benefit the few-shot classification under PFL scenarios due to its advantage of transferability, as shown in our experiments.

In view of this insight, our goal is to learn a set of personalized models $\{\theta_1^*, \ldots, \theta_n^*\}$ for n participating clients, each of which can produce a discriminative feature space by encouraging more similar clients to foster collaboration. More specifically, following the general FL setup [1], [31], pFedFSL iteratively performs the following three steps: 1) a client receives a set of clients' models from the FL server and then computes a weighted combination of the performance of the models on its own local data; 2) each client optimizes its

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model using its local data and then sends the optimized model and aggregation weights to the FL server; and 3) the FL server decides which subset of client models needs to be sent to a target client according to a sampling scheme. The three phases (one communication round) are presented in the following.

1) Model Aggregation Phase: Unlike the typical FL setup [1], [9], where model aggregation is performed in the FL server, pFedFSL conducts model aggregation for each client in a decentralized manner. This choice is due to the unknown distributions of local data; we can then perform model aggregation locally at clients according to their performance on local data. Suppose that client c_i has received a set of models $\{\theta^t_{n_1}, \ldots, \theta^t_{n_m}\}$ from the server. pFedFSL first computes personalized weights for the received models $\{\theta^t_{n_1}, \ldots, \theta^t_{n_m}\}$ and the local model θ^t_i and then uses these weights to fulfill model aggregation. Specifically, it is expected that a model $\theta^t_{n_j}$ has a high weight for model aggregation when it performs well on the local data $\mathcal{D}_i$, which may be caused by $\mathcal{D}_{n_j}$ having a similar data distribution as $\mathcal{D}_i$ or model $\theta^t_{n_j}$ being more fit for $\mathcal{D}_i$. Given this, we use a softmax function to compute the aggregation weights as follows:

$$w^t_{i,n_j} = \frac{\exp\left(-\mathcal{L}\left(\theta^t_{n_j}; \mathcal{D}_i\right)\right)}{\sum_{k=1}^{m+1} \exp\left(-\mathcal{L}\left(\theta^t_{n_k}; \mathcal{D}_i\right)\right)}, \quad \text{where } \theta^t_{n_{m+1}} = \theta^t_i \tag{1}$$

where w^t_{i,n_j} denotes the weight of model parameters $\theta^t_{n_j}$ for model aggregation in round t of client c_i, and it is calculated using the performance of model parameters $\theta^t_{n_j}$ on $\mathcal{D}_i$. Using the personalized set of weights $\{w^t_{i,n_1}, \ldots, w^t_{i,n_m}, w^t_{i,n_{m+1}}\}$, pFedFSL performs model aggregation for client c_i as

$$\widetilde{\theta}^t_i = \sum_{j=1}^{m+1} w^t_{i,n_j} \theta^t_{n_j} \tag{2}$$

where $\widetilde{\theta}^t_i$ is the aggregated model for c_i in round t. Then, pFedFSL takes this aggregated model $\widetilde{\theta}^t_i$ as a new starting point to perform local update with the aim of producing a more representative feature space for client c_i in round t.

2) Local Updating Phase: Inspired by prototypical networks [24], pFedFSL learns, for each client c_i with few-shot data $\mathcal{D}_i$, a personalized discriminative feature space produced by a network $f_{\theta_i} : \mathbb{R}^d \rightarrow \mathbb{R}^h$. Via f_{θ_i}, client c_i learns a new feature space that maps samples of the same category close to one another and samples of different categories far apart. This is also driven from the fact that the feature space embedding learned from one task can also perform well on other related tasks as it has the virtue of transferability and thus can eventually benefit other related tasks.

Given a sample $\mathbf{x}^i_j$ of client c_i, pFedFSL estimates the probability of label y^i_j as a softmax of the average distance of $\mathbf{x}^i_j$ from the samples of category y^i_j in the new feature space

$$p_{\widetilde{\theta}^t_i}\left(y^i_j \mid \mathbf{x}^i_j\right) = \frac{\exp\left(-\frac{1}{|\mathcal{S}^i_{y_j}|} \sum_{\mathbf{x} \in \mathcal{S}^i_{y_j}} d\left(f_{\widetilde{\theta}^t_i}\left(\mathbf{x}^i_j\right), f_{\widetilde{\theta}^t_i}(\mathbf{x})\right)\right)}{\sum_{l=1}^{N_i} \exp\left(-\frac{1}{|\mathcal{S}^i_{y_l}|} \sum_{\mathbf{x} \in \mathcal{S}^i_{y_l}} d\left(f_{\widetilde{\theta}^t_i}\left(\mathbf{x}^i_j\right), f_{\widetilde{\theta}^t_i}(\mathbf{x})\right)\right)} \tag{3}$$

where $d(f_{\widetilde{\theta}^t_i}(\mathbf{x}^i_j), f_{\widetilde{\theta}^t_i}(\mathbf{x})))$ is the Euclidean distance between samples $\mathbf{x}^i_j$ and $\mathbf{x}$ in the new feature space projected by $\widetilde{\theta}^t_i$. In doing so, pFedFSL achieves improved robustness compared to existing PFL and global FL methods when facing universal FL scenarios, where local clients have different label spaces. Current PFL methods [17], [18], [31] typically use classification layers with the same structure for all clients to perform prediction, which leads to error-prone models when local clients have different label spaces. To obtain a discriminative feature space for samples, pFedFSL minimizes the negative log-probability of the ground-truth label of sample $\mathbf{x}^i_j$ as follows:

$$J\left(\widetilde{\theta}^t_i, \mathbf{x}^i_j\right) = -\log\left(p_{\widetilde{\theta}^t_i}\left(y^i_j \mid \mathbf{x}^i_j\right)\right) \tag{4}$$

where $J(\widetilde{\theta}^t_i, \mathbf{x}^i_j)$ denotes the loss of model $\widetilde{\theta}^t_i$ on sample $\mathbf{x}^i_j$. Next, pFedFSL optimizes the model parameters by minimizing the average negative log probability of ground-truth labels of all training local samples and computes the loss $\mathcal{L}(\widetilde{\theta}^t_i; D_i)$ of client c_i as

$$\mathcal{L}\left(\widetilde{\theta}^t_i; D_i\right) = \frac{1}{N_i \cdot K_i} \sum_{j=1}^{N_i \cdot K_i} J\left(\widetilde{\theta}^t_i, \mathbf{x}^i_j\right) \tag{5}$$

where N_i and K_i denote the number of classes and the number of training samples per class, respectively, of the ith client. By minimizing (5), pFedFSL obtains a new feature space personalized by $\widetilde{\theta}^t_i$ for each client.

However, due to the few-shot samples of each client, the minimization of (5) leads to an overfit model. To overcome this issue, a sampling strategy is introduced in the FL server to adaptively send a set of model parameters to the current client with the aim of promoting its generalization ability. Unfortunately, sharing parameters among clients can compromise data privacy caused by malicious clients with inference attacks [43], [44]. As Zhao *et al.* [45] proved that sharing fewer parameters between clients and server can alleviate the privacy leakage, our pFedFSL induces a sparse model to address the privacy issue by compressing models and reducing the number of nonzero parameters of the models since compression can anonymize the data and preserve privacy by limiting the amount of information that a model shares [46]. Sparsity can be achieved in different ways, e.g., by adding a dropout layer in deep learning models or by adding an l_1-norm regularization. In pFedFSL, we use the l_1-norm regularization to encourage a sparse model, which is also robust against overfitting. The final loss function becomes

$$\widetilde{\mathcal{L}}\left(\widetilde{\theta}^t_i; \mathcal{D}_i\right) = \mathcal{L}\left(\widetilde{\theta}^t_i; \mathcal{D}_i\right) + \lambda \left\|\widetilde{\theta}^t_i\right\|_1 \tag{6}$$

where λ is the regularization parameter. With the target objective $\widetilde{\mathcal{L}}(\widetilde{\theta}^t_i; \mathcal{D}_i)$ in (6) and the local data $\mathcal{D}_i$, pFedFSL directly updates the model parameters through gradient descent as follows:

$$\theta^{t+1}_i = \widetilde{\theta}^t_i - \eta \nabla_{\widetilde{\theta}^t_i} \widetilde{\mathcal{L}}\left(\widetilde{\theta}^t; \mathcal{D}_i\right) \tag{7}$$

where η is the learning rate. Each client c_i sends the updated local model θ^{t+1}_i and personalized aggregation weights $\{w^t_{i,n_1}, \ldots, w^t_{i,n_m}, w^t_{i,n_{m+1}}\}$ to the FL server after accomplishing the local update. In addition, to further alleviate privacy leak issue, we can implement pFedFSL with the identification

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connection between clients and their uploaded models being private to other clients.

3) Selecting and Sharing Local Models: After receiving the model parameters and aggregation weights from all clients, the FL server selectively sends a subset of the received models to a target client by first assessing which models most benefit the target client. If the FL server simply sends all received models to each client, an overwhelming communication overhead would be raised. As an alternative, the FL server could randomly sample a subset of model parameters to be sent to a client. However, due to the fact that the data among clients are non-i.i.d., randomly sampling may not converge. As such, pFedFSL uses a sampling framework aimed at estimating which models most benefit a target client and then sends helpful models as many as possible to the target client.

Let $\mathbf{Q} \in \mathbb{R}^{n\times n}$ be an affinity matrix, where $\mathbf{Q}_{ij}$ measures how well θ_j performed on the data of client c_i in previous rounds. At each round, pFedFSL sends each client c_i the parameters of the m models corresponding to the top m values in the ith row of $\mathbf{Q}$. We initially set $\mathbf{Q}^1 = \text{diag}(-\text{inf}, \ldots, -\text{inf})$, where inf denotes infinity; as such, at the beginning, each model has an equal chance of being sent to client c_i except model θ_i. Then, at round t, $\mathbf{Q}^t_{ij}$ is updated using the weight $w^t_{i,j}$ calculated during the aggregation phase as follows:

$$\mathbf{Q}^{t+1}_{ij} = \begin{cases} \mathbf{Q}^t_{ij} + w^t_{i,j} - w^t_{i,n_{m+1}}, & \text{if } \theta^t_j \text{ is sent to } c_i \\ \mathbf{Q}^t_{ij}, & \text{otherwise} \end{cases} \tag{8}$$

where $w^t_{i,n_{m+1}}$ $(w^t_{i,i})$ is a threshold to determine whether θ^t_j performs well on $\mathcal{D}_i$. In this way, pFedFSL encourages the clients with similar data distribution to collaborate and induces a personalized model for each participating client with few-shot data. A client model may obtain a low weight in the initial phase due to trained with only few training epochs, and this model cannot fit the local data well and provide a good performance. As a result, it will be excluded from the subsequent aggregations and not broadcast to other clients, even if this client's task is similar to other clients. To tackle this "starve" problem, we adopt an R-delayed strategy and update $\mathbf{Q}$ after the first R communication rounds, during which the local model can fit the local distribution.

Algorithm 1 summarizes the three phases of pFedFSL: model aggregation (steps 4 and 5), local updating (steps 6–8), and selecting and sharing models (steps 10–14). We further analyze the communication and computational cost as follows.

1) *Communication Cost:* Compared with FedAvg, our pFedFSL additionally sends m local models' parameters to a target client and sends back a personalized local model and weight vector $\mathbf{w}_i \in \mathbb{R}^{m+1}$ to the FL server; as such, the extra communication in each round is $(m-1)CL(\theta) + CL(\mathbf{w}_i)$, where $CL(*)$ denotes the communication load of $*$. Since our model is sparse, the actual communication cost can be further reduced. In fact, we empirically find that an effective collaboration scheme can be achieved with $m \leq 5$ and pFedFSL converges much faster than FedAvg.
2) *Computational Cost:* Compared with other FL methods, our extra cost is to compute the loss of the received model for clients, in which the most time-consuming part is to calculate the distance between training samples. However, this extra cost is neglectable, due to scarce few-shot samples in each client.

Algorithm 1 pFedFSL

Input: n clients, where each client c_i carries local data $\mathcal{D}_i$ and randomly initialized model parameters θ_i; $T = 500$ (number of total communication rounds); $R = 25$ (delayed parameter); $\mathbf{Q} = \text{diag}(-\text{inf}, \cdots, -\text{inf})$; local test sample $\mathbf{x}^i$ on client i.
Output: Trained personalized model parameters $\{\theta^*_1, \cdots, \theta^*_n\}$.

1: *Server* sends m models' parameters corresponding to the top m values in the i-th row of $\mathbf{Q}$ to each c_i.
2: **for** $t = 1 \rightarrow T$ **do**
3: **for** all *clients* $i = 1 \rightarrow n$ **in parallel do**
4: Compute weights $\{w^t_{i,n_1}, \cdots, w^t_{i,n_{m+1}}\}$ via Eq. (1).
5: Conduct local model aggregation via Eq. (2).
6: Compute local loss via Eq. (6).
7: Optimize θ^t_i via Eq. (7).
8: Send θ^{t+1}_i, $\{w^t_{i,n_1}, \cdots, w^t_{i,n_{m+1}}\}$ to server.
9: **end for**
10: **for** *server* **do**
11: Receive θ^{t+1}_i, $\{w^t_{i,n_1}, \cdots, w^t_{i,n_{m+1}}\}$ from each c_i.
12: Update matrix $\mathbf{Q}$ via Eq. (8) when $t > R$.
13: Send m models' parameters corresponding to the top m values in the i-th row of $\mathbf{Q}$ to each c_i.
14: **end for**
15: **end for**
16: Classify $\mathbf{x}^i$ using Eq. (3) and θ^*_i.

The threat model, i.e., the malicious clients use model poisoning attack [47] to send arbitrary messages to undermine the training process, can be largely prevented by pFedFSL, which conducts model aggregation and model selection based on the performance of client models. The threat models will be assigned with low weights for aggregation and excluded from subsequent collaborations due to their poor performance.

IV. Experimental Results and Analysis

A. Experimental Setup

1) Datasets: We conduct experiments on four benchmark datasets MNIST [48], CIFAR-10 [49], CIFAR-100 [49], and miniImageNet [50]. For each dataset, we apply two different scenarios for simulating non-i.i.d. and few-shot data distributions across clients.

1) Pathological [1] non-i.i.d. and few-shot setup, where each client c_i is randomly allocated N_i classes and $K_i + 2K_i$ samples per selected category. Specifically, K_i samples and another $2K_i$ samples per selected category are randomly selected without replacement for each client to train and test, respectively. For each client, the number of classes N_i is randomly selected from the set $\{N-1, N, N+1\}$, and the number of samples per category for training K_i is randomly selected from the set $\{K-2, K-1, K, K+1, K+2\}$. In this way, the number of classes N_i and the number of samples per

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class $K_i + 2K_i$ are not the same across clients, i.e., each client c_i has $N_i \cdot K_i$ and $N_i \cdot 2K_i$ samples for training and testing, respectively.

2) Realistic non-i.i.d. and few-shot setup, where n participating clients are first allocated to five clusters (each cluster has $n/5$ different clients). Then, each cluster is randomly allocated N classes and $K \cdot n/5$ samples per chosen class for training. Next, for each cluster, the Dirichlet distribution with a distribution hyperparameter α is utilized to allocate total $N \cdot K \cdot n/5$ training samples to corresponding clients, where a smaller α indicates a more nonuniform distribution, and we randomly chosen other samples twice the number of training samples for each client to test. Following [51], we set $\alpha = 1$ in all experiments. We will study the impact of α later.

In the pathological setting, all clients are with different tasks due to the distinct target label spaces, while in the realistic setting, a subset of clients share the same target label space but with a different number of samples per category. The realistic setup is more practical than the pathological setting. In real-world scenarios, the number of samples varies significantly across the labels and participating clients, and not all the clients have distinct label spaces.

2) Compared Methods: To perform a comprehensive comparison, we compare pFedFSL against two typical global FL methods (FedAvg [1] and FedProx [9]), four recent and representative PFL methods (Per-FedAvg [7], pFedMe [17], HeurFedAMP [18], and FedFomo [31]), and FedFSL-adv [30] for federated FSL. Each compared method is configured with suggested parameters in the corresponding literature.

1) FedAvg [1] is a global FL approach, which performs model aggregation in server by averaging the models sent from clients.
2) FedProx [9] adds a proximal term to the local update to limit the impact of variable local updates, which boosts convergence behavior. Suggested configuration: regularization parameter $\mu = 0.01$.
3) Per-FedAvg [7] induces a shared global model that performs well on most clients within the MAML framework and then personalizes the learned global model using the client private dataset. Suggested configuration: second learning rate $\beta = 0.001$.
4) pFedMe [17] formulates the PFL problem as a bilevel optimization problem by decoupling the training of personalized models from inducing the global model. Suggested configuration: the regularization parameter $\lambda = 15$, the number of training steps for approximately finding the personalized model $K = 5$, and the global model moving parameter $\beta = 1$.
5) HeurFedAMP [18] maintains a personalized cloud model for each client on the FL server and then passes the personalized models from clients to personalized cloud models with similar model parameters. Suggested configuration: the regularization parameter $\lambda = 15$.
6) FedFomo [31] learns optimal weighted model combinations for each participating client by determining how much a client can benefit from other client models. Suggested configuration: the number of models sent to a client $m = 5$.
7) FedFSL-Adv [30] achieves the federated FSL in an adversarial fashion. Suggested configuration: adaptation step size $\alpha = 0.01$ and the loss tradeoff parameters $\eta = \lambda = 0.1$.

As to our pFedFSL, the l_1-norm regularization parameter $\lambda = 0.0005$, the delayed parameter $R = 25$, and the number of models sent for each client $m = 5$. In addition, for a fair comparison, all compared methods use the following configurations: the optimizer SGD, the learning rate $\eta = 0.01$, the number of participating clients in each communication round 5, the local update epochs $epoch = 5$, and the neural network proposed in [50] as the backbone, and it consists of four convolutional blocks, each of which is a 64-filter 3×3 convolution followed by a batch normalization layer, a ReLU nonlinearity, and a 2×2 max-pooling layer. The number of clients n is set to 30 for all methods. For each method, we report the mean and standard deviation of the accuracy of five independent rounds, where in each round, the average accuracy of all clients is recorded.

B. Results and Analysis

We conduct experiments on two non-i.i.d. and few-shot settings (pathological and realistic) with the following controls: $N \in \{3, 5\}$ and $K \in \{5, 20\}$ on MNIST and CIFAR-10, respectively, and $N \in \{5, 20\}$ and $K \in \{5, 20\}$ on more large-scale CIFAR-100 and miniImageNet, respectively. The experimental results are reported in Table I. From this table, we have the following observations.

1) pFedFSL outperforms the other methods across all the settings, which proves the effectiveness of pFedFSL on two non-i.i.d. and few-shot scenarios. As N steps from 3 to 5 (MNIST and CIFAR-10) or from 5 to 20 (CIFAR-100 and miniImageNet) under a fixed K, all methods have a decreased performance except for global FL methods (FedAvg and FedProx), which manifests an improved performance. This is due to the increased class labels and task complexity (the random guess accuracy decrease from 1/3 to 1/5 or from 1/5 to 1/20), and the global FL methods only induce one model to fit all local data, which benefits more from increased training samples (the training samples increase from $3K$ to $5K$ or from $5K$ to $20K$ in each client). On the other hand, as K increases under a fixed N and dataset, almost each method shows an increasing accuracy due to more samples for training. The performance gap between pFedFSL and the compared methods is more prominent when K is small since these methods build on the premise that most or all clients have sufficient samples for training. This result confirms the impact of few-shot data on the PFL problem and also proves the superiority of pFedFSL. Moreover, each method has a better performance on MNIST than on miniImageNet, and this is because MNIST is a grayscale image dataset with only ten labels. In contrast, miniImageNet is a color image dataset with 100 labels.

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TABLE I

CLASSIFICATION ACCURACY (MEAN ± STD) OF COMPARISON METHODS ON TWO NON-I.I.D. AND FEW-SHOT SETTINGS. $N(K)$: NUMBER OF CLASSES (TRAINING SAMPLES PER CLASS) FOR EACH CLIENT. THE BEST PERFORMANCE IN EACH SETTING IS BOLD-FACED

	Pathological				Realistic			
	$N=3$		$N=5$		$N=3$		$N=5$	
	$K=5$	$K=20$	$K=5$	$K=20$	$K=5$	$K=20$	$K=5$	$K=20$
	MNIST							
FedAvg	0.885±0.010	0.943±0.005	0.913±0.006	0.960±0.002	0.908±0.017	0.963±0.005	0.917±0.010	0.955±0.002
FedProx	0.893±0.006	0.948±0.006	0.910±0.007	0.958±0.004	0.918±0.010	0.962±0.007	0.920±0.001	0.957±0.005
Per-FedAvg	0.883±0.011	0.947±0.006	0.904±0.010	0.949±0.005	0.916±0.009	0.952±0.010	0.925±0.006	0.955±0.003
pFedMe	0.919±0.016	0.943±0.005	0.907±0.006	0.936±0.002	0.926±0.009	0.961±0.006	0.927±0.009	0.958±0.006
HeurFedAMP	0.878±0.014	0.956±0.004	0.785±0.015	0.930±0.008	0.871±0.017	0.942±0.009	0.847±0.019	0.915±0.006
FedFomo	0.916±0.015	0.971±0.007	0.909±0.004	0.957±0.003	0.934±0.017	0.970±0.006	0.918±0.013	0.954±0.006
FedFSL-Adv	0.908±0.006	0.966±0.005	0.916±0.009	0.954±0.010	0.929±0.014	0.962±0.016	0.927±0.018	0.956±0.015
pFedFSL	**0.933±0.007**	**0.976±0.002**	**0.931±0.009**	**0.975±0.003**	**0.952±0.012**	**0.985±0.002**	**0.948±0.009**	**0.976±0.006**
	CIFAR-10							
FedAvg	0.335±0.015	0.446±0.028	0.380±0.019	0.482±0.007	0.425±0.045	0.529±0.031	0.426±0.024	0.485±0.025
FedProx	0.355±0.008	0.433±0.016	0.390±0.009	0.489±0.020	0.404±0.054	0.517±0.039	0.425±0.025	0.505±0.020
Per-FedAvg	0.426±0.017	0.508±0.016	0.388±0.015	0.455±0.020	0.478±0.027	0.594±0.009	0.440±0.045	0.554±0.019
pFedMe	0.522±0.038	0.606±0.035	0.420±0.021	0.539±0.010	0.590±0.024	0.672±0.024	0.521±0.032	0.571±0.020
HeurFedAMP	0.564±0.031	0.657±0.022	0.377±0.018	0.511±0.012	0.660±0.021	0.742±0.009	0.506±0.018	0.619±0.019
FedFomo	0.565±0.039	0.679±0.024	0.428±0.017	0.538±0.014	0.669±0.024	0.756±0.016	0.521±0.012	0.621±0.025
FedFSL-Adv	0.532±0.039	0.635±0.029	0.397±0.018	0.555±0.026	0.566±0.029	0.697±0.038	0.472±0.024	0.624±0.028
pFedFSL	**0.604±0.013**	**0.702±0.012**	**0.440±0.024**	**0.580±0.020**	**0.694±0.025**	**0.765±0.025**	**0.549±0.015**	**0.668±0.018**
	$N=5$		$N=20$		$N=5$		$N=20$	
	$K=5$	$K=20$	$K=5$	$K=20$	$K=5$	$K=20$	$K=5$	$K=20$
	CIFAR-100							
FedAvg	0.107±0.010	0.175±0.017	0.130±0.015	0.229±0.012	0.310±0.037	0.511±0.021	0.221±0.015	0.269±0.032
FedProx	0.116±0.013	0.173±0.021	0.136±0.007	0.227±0.006	0.341±0.027	0.463±0.038	0.216±0.020	0.302±0.043
Per-FedAvg	0.216±0.032	0.267±0.018	0.172±0.008	0.231±0.008	0.427±0.036	0.531±0.048	0.238±0.024	0.327±0.013
pFedMe	0.338±0.015	0.416±0.014	0.204±0.014	0.274±0.014	0.531±0.027	0.572±0.033	0.303±0.016	0.335±0.014
HeurFedAMP	0.462±0.028	0.596±0.009	0.211±0.011	0.305±0.005	0.555±0.030	0.693±0.026	0.304±0.013	0.426±0.012
FedFomo	0.479±0.022	0.600±0.007	0.222±0.009	0.311±0.008	0.582±0.022	0.697±0.027	0.326±0.015	0.431±0.024
FedFSL-Adv	0.230±0.035	0.395±0.022	0.219±0.014	0.276±0.011	0.453±0.044	0.659±0.043	0.307±0.006	0.361±0.022
pFedFSL	**0.488±0.022**	**0.641±0.017**	**0.232±0.004**	**0.394±0.003**	**0.616±0.021**	**0.726±0.024**	**0.342±0.018**	**0.461±0.007**
	miniImageNet							
FedAvg	0.077±0.007	0.114±0.009	0.093±0.005	0.159±0.005	0.232±0.029	0.377±0.050	0.154±0.017	0.198±0.031
FedProx	0.083±0.021	0.112±0.018	0.088±0.008	0.151±0.004	0.241±0.033	0.366±0.037	0.145±0.022	0.194±0.025
Per-FedAvg	0.230±0.020	0.462±0.027	0.114±0.016	0.189±0.014	0.317±0.032	0.480±0.026	0.186±0.009	0.251±0.023
pFedMe	0.299±0.012	0.381±0.022	0.160±0.009	0.223±0.012	0.456±0.026	0.526±0.031	0.241±0.010	0.285±0.006
HeurFedAMP	0.401±0.012	0.539±0.017	0.160±0.010	0.248±0.005	0.521±0.018	0.625±0.021	0.258±0.009	0.342±0.014
FedFomo	0.411±0.013	0.526±0.010	0.169±0.005	0.245±0.012	0.549±0.025	0.646±0.021	0.265±0.011	0.354±0.010
FedFSL-Adv	0.158±0.020	0.346±0.023	0.142±0.016	0.211±0.019	0.420±0.039	0.560±0.042	0.232±0.025	0.279±0.040
pFedFSL	**0.432±0.031**	**0.559±0.009**	**0.183±0.008**	**0.303±0.011**	**0.564±0.017**	**0.652±0.015**	**0.271±0.013**	**0.391±0.009**

2) *Pathological Versus Realistic:* Each method has a higher performance under the realistic scenario in most cases. This is because the tasks among clients under the pathological scenario (all clients are with distinct tasks) are more heterogeneous than those under the realistic scenario (a subset of clients are with the same tasks).

3) *pFedFSL Versus Global FL and PFL:* The global FL methods (FedAvg and FedProx) generally lose to the PFL methods, which signifies that a single global model cannot fit well all non-i.i.d. local data. This also proves the effectiveness of PFL methods under non-i.i.d. scenarios. However, these PFL methods still lose to pFedFSL, especially with respect to a small K, since they build on the premise that most or all clients have sufficient samples for training. FedFSL-Adv also depends on the composition of many base tasks to train clients' FSL models. As such, it is outperformed by pFedFSL across the two settings as well.

C. Further Analysis

1) Different Levels of Heterogeneity: We further conduct experiments on MNIST to study the performance of pFedFSL and the competitive methods under the realistic scenario with different levels of heterogeneity by varying the Dirichlet distribution hyperparameter α. Table II reports the accuracy of each compared method as $N = 5$ and $K = 5$. All compared methods have an improved performance as α increases. This is due to the lower distribution heterogeneity among clients with

TABLE II

ACCURACY (MEAN ± STD) OF COMPARISON METHODS ON MNIST UNDER REALISTIC SCENARIO AS $N = 5$ AND $K = 5$ WITH DIFFERENT LEVELS OF HETEROGENEITY. α: DIRICHLET DISTRIBUTION HYPERPARAMETER, AND A SMALLER VALUE INDICATES A LARGER HETEROGENEITY. THE BEST PERFORMANCE IN EACH SETTING IS BOLD FONT

	$\alpha=1$	$\alpha=10$	$\alpha=100$	$\alpha=1000$
FedAvg	0.917±0.010	0.937±0.008	0.939±0.008	0.945±0.009
FedProx	0.920±0.001	0.941±0.008	0.937±0.005	0.941±0.010
Per-FedAvg	0.925±0.006	0.937±0.008	0.942±0.005	0.937±0.012
pFedMe	0.927±0.009	0.931±0.005	0.936±0.004	0.937±0.008
HeurFedAMP	0.847±0.019	0.867±0.008	0.888±0.017	0.872±0.027
FedFomo	0.918±0.013	0.930±0.019	0.933±0.012	0.937±0.011
FedFSL-Adv	0.927±0.018	0.931±0.008	0.944±0.015	0.943±0.011
pFedFSL	**0.948±0.009**	**0.950±0.015**	**0.955±0.013**	**0.950±0.011**

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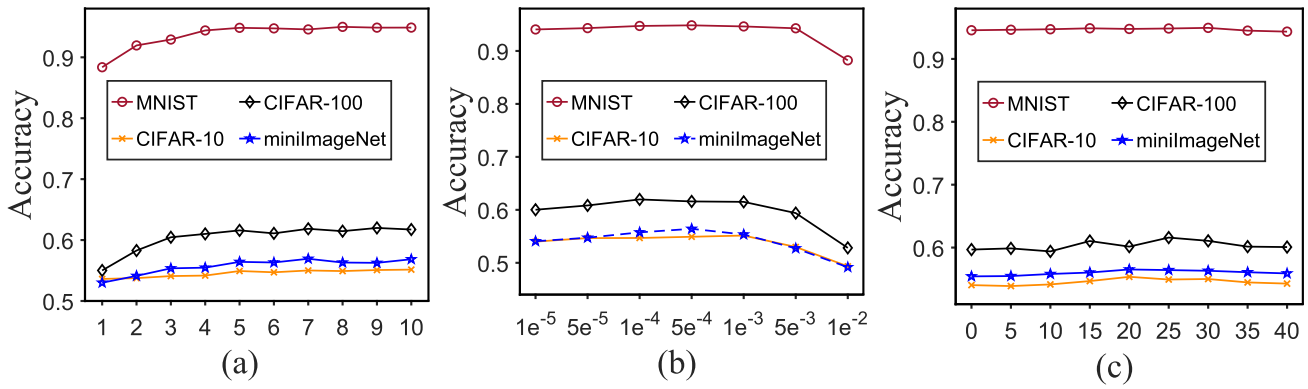

Fig. 2. Performance of pFedFSL with various parameters with $N = 5$ and $K = 5$ in the realistic setting: (a) accuracy versus m, (b) accuracy versus λ, and (c) accuracy versus R.

TABLE III

PERFORMANCE OF pFEDFSL AND ITS DEGENERATED VARIANTS UNDER THE REALISTIC NON-I.I.D. SCENARIO WITH $N = 5$ AND $K = 5$

	MNIST	CIFAR-10	CIFAR-100	miniImageNet
pFedFSL-nF	0.793±0.017	0.493±0.019	0.573±0.020	0.510±0.018
pFedFSL-nR	0.933±0.009	0.535±0.016	0.597±0.012	0.538±0.017
pFedFSL-nS	0.935±0.006	0.529±0.014	0.596±0.014	0.542±0.015
pFedFSL	**0.948±0.009**	**0.549±0.015**	**0.616±0.021**	**0.564±0.017**

a bigger α. This fact not only signifies the impact of statistical heterogeneity of non-i.i.d. local data but also corroborates the necessity to account for such heterogeneity. In addition, the global FL methods outperform the PFL methods under a big α since the lower statistical heterogeneity of clients' data and a global model fits multiple clients well. However, they still lose to our pFedFSL since pFedFSL can learn a personalized and discriminative feature space for each client and depends less on the number of training samples than other methods.

2) Ablation Study: We conduct an ablation study to further analyze the contribution factors of pFedFSL. For this purpose, we introduce three variants of pFedFSL: pFedFSL-nF, pFedFSL-nR, and pFedFSL-nS. Specifically, pFedFSL-nF independently trains the personalized model for each client without FL, pFedFSL-nR does not use the l_1-norm regularization for sparse model, and pFedFSL-nS randomly sends m models to each client, without adaptive selecting models. Table III gives the performance of three variants and pFedFSL under the realistic non-i.i.d. scenario with $N = 5$ and $K = 5$. We observe that the following conditions hold.

1) The big performance gap between pFedFSL-nF and pFedFSL proves the necessity and effectiveness of FL.
2) The l_1-norm regularization helps pFedFSL alleviating the overfitting problem. This is verified by the clear margin between pFedFSL and pFedFSL-nR. Besides preventing the model from being overfit, the l_1-norm can also protect privacy by pursuing sparse local models. In fact, we find that the parameter size of sparse model is 65.7% of the nonsparse counterpart, which greatly reduces the communication load.
3) The fact that pFedFSL has a better performance than pFedFSL-nS demonstrates the effectiveness of the adaptive strategy in selecting a subset of models for clients, which can encourage useful collaboration among clients and alleviate data scarcity.

3) Parameter Analysis: We study the parameter sensitivity of pFedFSL with respect to m (number of sending models), λ (regularization parameter), and R (delayed parameter) by varying one parameter while fixing other ones to our finally used values under the realistic scenario with $N = 5$ and $K = 5$ and reveal the results in Fig. 2. We have the following observations.

1) *Sensitivity to m:* Fig. 2(a) reports the accuracy of pFedFSL as m varies in the range $\{1, 2, \ldots, 10\}$. pFedFSL has an improved performance as m increases until $m \approx 5$, which again proves the effectiveness of FL. Indeed, it also indicates that an effective collaboration can be achieved with a small $m \leq 5$, which further limits the communication overhead.
2) *Sensitivity to λ:* Fig. 2(b) reports the accuracy of pFedFSL as λ varies in the range $\{1e^{-5}, 5e^{-5}, \ldots, 1e^{-2}\}$. pFedFSL first manifests an increased accuracy as λ increases until $\lambda \approx 5e^{-4}$, which again demonstrates the effectiveness of l_1-norm regularization. However, the accuracy begins to decrease as λ further increases, due to the overweight of l_1-norm regularization, which leads to a too sparse model to be effective.
3) *Sensitivity to R:* Fig. 2(c) reports the accuracy of pFedFSL as R varies in the range $\{0, 5, \ldots, 40\}$. pFedFSL has a slight improvement on performance as R increases until $R \approx 25$ and then has a minor reduced performance as R further raises. This is because the R-delayed update strategy can alleviate the "starve" problem of client models, and the update of $\mathbf{Q}$ in (8) enables effective collaboration among client models.

Based on the above observations, pFedFSL adopts $m = 5$, $\lambda = 5e^{-4}$, and $R = 25$ for experiments.

4) Impact of the Number of Clients: Fig. 3 reports the performance of each compared method on MNIST as a number of participating clients n steps from 10 to 100 under realistic scenario on MNIST with $N = 5$ and $K = 5$. Each method has an improved performance as n increases since more knowledge

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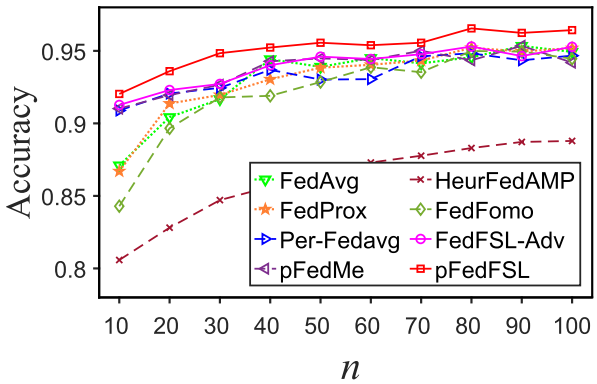

Fig. 3. Accuracy versus number of clients n on MNIST with $N = 5$ and $K = 5$ in the realistic setting.

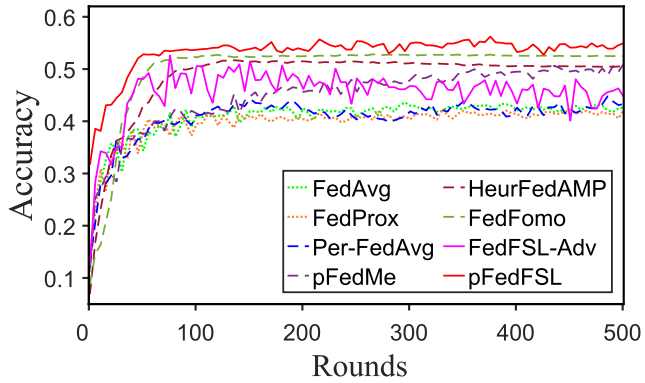

Fig. 4. Accuracy versus communication rounds on CIFAR with $N = 5$ and $K = 5$ in the realistic setting.

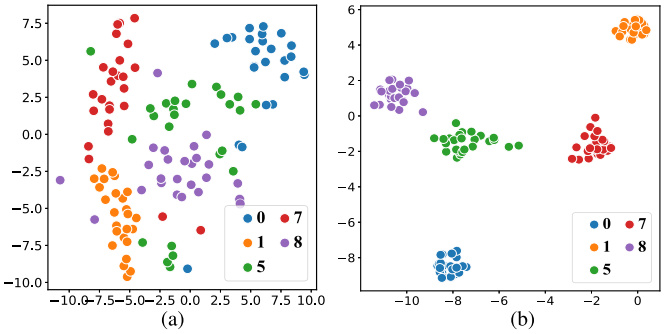

Fig. 5. Visualization of (a) original and (b) new feature spaces of training and testing samples owned by a client under pathological scenario with $N = 5$ and $K = 5$ on MNIST.

from clients can be leveraged to boost the performance; otherwise, pFedFSL still has the best performance with various n's, which again demonstrates the effectiveness of our pFedFSL.

5) Convergence Rate: Fig. 4 reports the convergence rate of each compared method under realistic scenario on CIFAR-10 with $N = 5$ and $K = 5$. pFedFSL reaches the maximum accuracy with about 120 communication rounds, and it is much faster than other methods except for HeurFedAMP and FedFomo, which separately apply heuristics distance approximation and first-order optimization to update local deep models. This further limits the communication overhead.

6) Visualization of New Feature Space: pFedFSL learns a personalized and discriminate feature space for each client. To verify the effectiveness of the learned feature space, we visualize the original/new feature space of a certain client (with class 0, 1, 5, 7, and 8) under pathological scenario on MNIST ($N = 5$ and $K = 5$) using t-SNE [52]. As shown in Fig. 5, compared with samples in the original space, the samples (K training samples plus $4K$ testing samples per category) of the same category are nearby while far away from samples of different categories in the new feature space. This merit enables training an accurate classifier and supports our motivation to perform collaboration between clients based on these new spaces.

V. Conclusion and Future Work

In this article, we focus on a practical and important but unexplored FL problem, where most or all participating clients only have a handful of few-shot training samples. Our proposed pFedFSL learns a personalized and discriminative feature space for each client and encourages more cooperation among similar clients. Extensive experiments confirm the efficacy of pFedFSL in different scenarios. However, the data heterogeneity exists not only in label space but also feature space and clients are often asynchronous in communication rounds; thus, we will extend our solution for FL with both dual and feature and label heterogeneity in an asynchronous scenario.

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